

Sean Zhao

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Education

University of Toronto

Toronto, ON

Bachelor of Applied Science in Computer Engineering

Expected Apr. 2029

- GPA: 3.97/4.00; Dean's Honour List; ECE Top Student Award
- Relevant coursework: Computer Fundamentals (C); Programming Fundamentals (C++ object-oriented programming, data structures and algorithms; in progress); Digital Systems (in progress)

Experience

Data Security Developer Intern

May 2026 – Aug. 2026

1Password

Toronto, ON

- Developed and tested production data security software in Rust, contributing from design through implementation
- Evaluated performance and compatibility across supported platforms during a security upgrade
- Received the Intern Trailblazer Award for innovation and adoption of new technologies

Projects

Pilot | *TypeScript, JavaScript, Playwright, MCP*

Sept. 2026

- Built a tool that lets apps and coding agents retrieve data from websites without official APIs through shared commands such as jobs.search
- Built a Playwright-based agent that generates and validates reusable JavaScript drivers; saved drivers run without new AI model calls, and Pilot repairs them after site changes
- Integrated drivers with Codex and Claude Code through Model Context Protocol (MCP); measured 50% fewer tokens in an app-building test using existing drivers than without Pilot

GameNet | *Rust, Tokio, QUIC, TLS 1.3, ML-KEM*

Jan. – Sept. 2026

- Built a TCP tunnel that lets friends join self-hosted game servers with normal clients; a public relay forwards each player to the host over a separate QUIC stream
- Secured the host-to-relay link with verified TLS 1.3 certificates and mandatory X25519MLKEM768 hybrid key exchange
- Added expiring port claims, connection limits, host-token rotation and theft recovery, with automated tests for failure paths
- Benchmarked synthetic TCP echo from the same EC2 client to a Mac host: 44–47 ms median open-connection reply via GameNet versus 213–215 ms via Pinggy, a similar TCP tunnel

Alias | *Next.js, FastAPI, Ghidra, LLM4Decompile, Gemini, Docker, Modal*

Jan. 2026

- Built a tool that analyzes executables without running them, showing readable C, a plain-English explanation of behavior, and potential security risks
- Combined Ghidra decompilation, LLM4Decompile refinement and Gemini analysis; deployed containerized inference on Modal

Awards

UofTHacks 13 (2026): Best Cybersecurity Hack (1Password sponsor track) for Alias

HackMIT 2026: The Token Company LLM Price-Saving Challenge winner for Pilot

CyberTitan Nationals (2025): 1st place overall

CyberPatriot National Finals (2025): Cisco 2nd place; 6th overall

Canadian Computing Competition: Junior perfect score (2023); Senior Distinction (2024, 2025)

Technical Skills

Languages: Rust, C, C++, Python, TypeScript, JavaScript, Java

Systems & Backend: Linux, Docker, Git, Tokio, QUIC, TLS 1.3, Node.js, Express.js, FastAPI

Web & Data: React, Next.js, PostgreSQL, MongoDB, OpenCV, MediaPipe

AI & Security: MCP, LLM driver generation and validation, applied cryptography, Ghidra